# Alif Azam

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## Research Interest

Computational Fourier Optics · Physics-Informed Neural Networks
Scientific Machine Learning · High-Performance Computing

## Skills

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|---|---|
| **Languages** | Python, C, Bash Scripting |
| **Packages and Libraries** | NumPy, Pandas, Matplotlib |
| **HPC Tools** | Slurm, OpenMPI, OpenMP, CUDA, Git |
| **Operating Systems** | Linux, Windows |

## Education

**B.Sc. Hons. in Physics**
Shahjalal University of Science and Technology, Sylhet — January 2023-Present

## Research Experience

**Nonlinear Optics Laboratory**
Department of Physics, SUST — December 2024 – Present

•Building a physics-informed spectral enhancement pipeline for label-free cell segmentation in phase-contrast microscopy, combining Fourier-domain quality assessment with PINN-based image restoration and adaptive bandpass filtering.

## Projects

**Full Diskless Linux HPC Cluster** 2025

•Built and deployed a 7-node diskless Linux cluster using PXE boot, DHCP, TFTP, NFS, MUNGE, and Slurm; configured the master node to provide network booting, shared root filesystem, authentication, and workload management for compute nodes.

•Implemented matrix multiplication with OpenMPI and OpenMP to test parallel execution and validate the cluster's distributed and shared-memory computing capabilities.
*full details*

**Nuclear Power Plant Data Analysis** 2025

•Analyzed ERA5 temperature data in NetCDF4 format using Python, working with a spatial subset covering Bangladesh and performing data extraction, preprocessing, visualization, and statistical analysis.
*full details*

# Publications

## Conference Proceedings

1. Nripendro Biswas, Ahmed Rijvee, Munaem Ahmed Mahdi, **Alif Azam**, Md. Takoat Hossain, *"BSER – Bengali Speech Emotion Recognition using MFCC Features Based on SUBESCO and BanglaSER: A Comparative Study of Machine Learning Models."* Proceedings of the 8th International Conference on Engineering Research, Innovation and Education (ICERIE 2025), Atlantis Press. DOI: 10.2991/978-94-6463-884-4_77

# Presentations

1. *"BSER - Bengali Speech Emotion Recognition using MFCC Features Based on SUBESCO and BanglaSER: A Comparative Study of Machine Learning Models"*: Nripendro Biswas, Ahmed Rijvee, Munaem Ahmed Mahdi, **Alif Azam** and Md. Takoat Hossain. *International Conference on Engineering Research, Innovation and Education (ICERIE 2025).*

# Linguistic Skill

English (Fluent) · Bangla (Native)

# Workshops/Courses

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|---|---|
| **ICTP** | *Invitation* |
| Advanced School in Applied Machine Learning: Uncertainty and Robustness of DL Models for Science (Remote participation) | July 2026 |
| **University of Colorado Boulder** | *Certificate* |
| Introduction to High-Performance and Parallel Computing | June 2026 |
| **Google** | *Certificate* |
| System Administration and IT Infrastructure Services | Jan 2025 |
| Google Digital Marketing Course | July 2021 |
| **SoloLearn** | *Certificate* |
| Python Core Course | Feb 2020 |